Course: MAT 112 – Linear Algebra and Differential Equations

Department of Software Engineering, Metropolitan University

6 questions will be given, 4 should be answered

Topics –

1. Complex Variables: (2 sets of Questions from this part)
    - Expressing into polar form. (Don't forget to check the position of a point in the quadrant, to determine the angle θ properly.)
    - Mathematical operations of complex numbers and graphical representation.
      Examples: 1.2, 1.4, 1.5, 1.7, 1.17, 1.43, 1.48, 1.116 (a, c)
    - De Moivre's Theorem: Statement and proof. Proof from other propositions. (As given in class test)
    - Roots of complex numbers: Example: 1.28, 1.29, 1.30, 1.37

2. Differential Equation:
    A. Ordinary Differential Equation (2.5 sets from this part)
        - Determine whether a function is a solution of a differential equation
          Example – 1.3-1.9, 1.13
        - Determining different types of D.E and Solution of those differential equations – (i) Variable Separable Equations (ii) Homogeneous ODE (iii) Linear ODE (iii) Exact ODE (all of first order.)
          ---Practice the problems solved in the class and given as homework---
          From Book:
          <u>Variable Separable and Homogenous Equation:</u> Schaum's Outlines: Page 22 – 4.1-4.3, 4.5-4.7, 4.10, 4.12, 4.14, 4.15)
          <u>Linear ODE and Exact ODE:</u> BD Sharma --- Page 18 of Part I: Ex-1, 2,3,10
          Page 37 of part I: Ex – 1, 3, 6
        - Linear differential equations with constant coefficients: (of Higher Order)
          <u>From BD Sharma:</u> Page 63 of Part I – 1,2,3.

    B. Partial Differential Equations (1.5 sets from this part)
        - <u>Linear PDE of Order one:</u> Lagrange's Method

From BD Sharma: Page 7 of Part III: Ex – 1,2,3

- Non-linear PDE: Charpit's Method
  From BD Sharma: Page 24 of Part III: Ex – 1,2
- Fourier Series
  Definition: Fourier Series, for even function, for odd function
  From S L Ross.: Page 613, 12.14, 12.15